Sidharth Singh

Backend engineer building high-throughput distributed systems and AI-powered platforms

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SUMMARY

Backend-leaning engineer with 2+ years of production experience building high-throughput microservices, event-driven pipelines and AI-assisted workflows for procurement, logistics and supply-chain platforms. Rust (Actix Web, Tokio) and TypeScript (NestJS) for services, Python (FastAPI) for data-heavy workflows, Kafka for everything asynchronous, PostgreSQL and MongoDB at 100M+ document scale. Owns a feature end to end, from marketplace or carrier integration through to production, and treats AI coding tools as part of design, implementation and review.

EXPERIENCE

Mar 2026 – Present
Okhla, Delhi

Software Engineer

1buy.ai · Sharpbuy Global Solutions Pvt. Ltd.

AI-powered B2B electronic component intelligence, sourcing and procurement platform.

- Design and run the order-fulfilment backend's architecture, built from scratch: a feature-first modular monolith in Python (FastAPI) shipped as one image in three roles (a 4-process API, N background workers, a leader-elected scheduler) over PostgreSQL as both source of truth and job queue, with SELECT … FOR UPDATE SKIP LOCKED claims, a stale-lease reclaim, a transactional outbox between bounded contexts and import-linter contracts in CI.
- Build the fulfilment workflows on it: purchase order to lab testing, shipment, customs and escrow, with DHL booking and tracking, FedEx tracking, and Claude parsing trade documents (packing lists, Bills of Entry, provider invoices).
- Own the inbound-mail pipeline: cheapest-first anchoring of every email to its order or shipment, a two-stage classify-then-act flow over a plug-in registry of 17 LLM handlers, per-step model routing, and dispatch through the outbox, with deterministic no-network tests.
- Design and build end-to-end backend services on a TypeScript, NestJS and MongoDB stack with hexagonal architecture, on an ultra-small engineering team.
- Engineer MongoDB aggregation pipelines and compound-indexing strategies over 100M+ pricing documents to power real-time component discovery.
- Run the background job scheduler I rewrote in Rust on Tokio: Arc<Semaphore>-bounded worker pools, atomic job claims and cancellation-safe tasks for concurrent, multi-instance processing without thread contention.
- Share pricing and formula logic between TypeScript and Rust through one Rust crate compiled to WebAssembly with wasm-bindgen, batches crossing a columnar Float64Array boundary, with parity tests on both sides.
- Develop automated RFQ ingestion pipelines and multi-currency calculation engines with Decimal128 precision, integrating LLM APIs into sourcing workflows.

`Python · FastAPI · PostgreSQL · Redis · Kafka · TypeScript · NestJS · MongoDB · Rust · Tokio · WebAssembly · Claude API`

Apr 2025 – Feb 2026
Gurugram

Software Engineer

RapidShyp · Om Logistics Limited

B2B logistics and shipping platform.

- Engineered core backend microservices in Rust (Actix Web on the Tokio runtime) and Java: async handlers, serde-typed domain models and backpressure-aware worker pools, improving API concurrency and throughput while reducing latency under peak shipping loads.
- Built event-driven pipelines on Apache Kafka for real-time order processing, shipment status synchronisation and multi-seller fulfilment across marketplace and storefront sales channels.
- Integrated carrier and logistics partner APIs for shipment creation, tracking and status updates, so sellers on multiple sales channels ship through one platform.
- Integrated Redis for low-latency caching alongside TiKV and etcd for distributed key-value storage, service coordination and configuration management.
- Optimised schema design and query execution across MongoDB and MySQL for high-volume logistics read and write paths.
- Won first place across all engineering teams in the RapidShyp internal hackathon (Mar 2025) and the Customer Obsession Award (May 2025) for feature ownership, rapid execution and reliable production delivery.

`Rust · Actix Web · Java · Kafka · MongoDB · MySQL · Redis · TiKV · etcd`

Jul 2024 – Mar 2025
Gurugram

Associate Software Engineer

RapidShyp · Om Logistics Limited

- Designed and owned the B2B Bulk Orders and Purchase Requisition flows in the Order Management System: bulk upload and validation of 20,000 orders in a single call, multi-seller shipment creation, approval workflows and order orchestration at scale.
- Brought marketplace and multi-channel seller orders into one fulfilment pipeline, so every sales channel flows through the same order lifecycle and logistics integrations.
- Developed order-lifecycle dashboards and the internal Warehouse Management System frontend in React and Next.js, used daily by operations teams.
- Contributed to RESTful microservices in Rust (Actix Web) and Java, integrating third-party logistics carrier APIs.
- Containerised local development environments with Docker and optimised query patterns across MongoDB and MySQL.

`Rust · Java · React · Next.js · MongoDB · MySQL · Docker`

EDUCATION

Sept 2020 – Jun 2024

B.Tech, Computer Science & Engineering (Bioinformatics)
Vellore Institute of Technology (VIT), Vellore
CGPA 8.22 / 10

SKILLS

Languages	`Rust · TypeScript · JavaScript · Python · Java · SQL · Go`
Backend & APIs	`Actix Web · Tokio · NestJS · FastAPI · Spring Boot · REST APIs · gRPC · WebSockets · Hexagonal architecture · Microservices`
Data & messaging	`PostgreSQL · MongoDB · MySQL · Redis · Apache Kafka · etcd · TiKV`
Infra & cloud	`Docker · Kubernetes · AWS (EC2, S3, Lambda) · CI/CD · Linux`
AI & search	`LLM integration (Claude) · RAG · Atlas Vector Search · pgvector · AI document parsing`
Frontend	`React · Next.js`

AWARDS & CERTIFICATIONS

Awards

2025 — **First place, RapidShyp internal hackathon 2025**
Across all engineering teams, for an innovative prototype built under tight time constraints (Mar 2025)

2025 — **Customer Obsession Award, RapidShyp**
For exceptional feature ownership, rapid execution, going the extra mile and reliable production delivery (May 2025)

Certifications

2023 — **Google Cloud Computing Foundations**
Google & VIT

2023 — **Modern Application Development with Java Spring Boot**
VMware IT Academy

2023 — **Core Technical Skills (Docker, Kubernetes, MongoDB)**
VMware IT Academy

SELECTED PROJECTS

D-MPNN drug property predictor

A directed message-passing neural network with ensemble learning for ADME property prediction, trained on 12,859 molecular SMILES entries with substructure features.

`Python · PyTorch · Cheminformatics · Ensemble learning`

Personal RAG assistant on AWS

A retrieval-augmented chat service built to learn the full build-and-deploy loop: NestJS, MongoDB Atlas vector search, Hugging Face embeddings, Groq-hosted Llama, Docker, and a hand-provisioned EC2 box.

`NestJS · TypeScript · MongoDB Atlas · Hugging Face · Groq · Docker`